Notes: Giroud and Mueller (QJE, 2017)

Firm Leverage, Consumer Demand, and Employment Losses During the Great Recession

Contents

1 Big picture

- **Question.** Do firm balance sheets amplify local consumer-demand shocks during the Great Recession? More precisely, when local house prices fall and household demand contracts, do establishments belonging to more highly levered firms cut employment more aggressively than otherwise similar establishments?
- **Setting.** The paper studies U.S. establishments between 2007 and 2009 by matching Census establishment-level employment data to parent-firm balance sheets and local house prices. The key comparison is between establishments facing similar local demand shocks but belonging to firms with different leverage.
- **Core challenge.** Falling house prices and local employment losses are jointly determined by local conditions. Moreover, highly levered firms may differ from low-leverage firms in many other ways: productivity, pre-crisis expansion, activist ownership, or general cyclical sensitivity. The paper therefore needs to isolate a balance-sheet interaction from these alternative channels.
- **Main result.** Establishments of more highly levered firms experience significantly larger employment losses in response to local house-price declines. In the baseline establishment-level regressions, the elasticity of employment with respect to local house prices rises strongly with parent-firm leverage. The implied employment sensitivity at the 90th percentile of leverage is about three times the sensitivity at the 10th percentile.
- **Sectoral result.** The interaction is concentrated in nontradable and "other" local-demand sectors, while there is no meaningful effect in the tradable sector. This is exactly what one would expect if the mechanism operates through local consumer demand rather than through generic firm weakness.
- **Financial-constraint mechanism.** More highly levered firms look and behave like financially constrained firms. When hit by negative local demand shocks, they are less able to raise additional short-term and long-term debt, close more establishments, and cut investment more sharply.
- **Aggregate result.** The effect survives aggregation. Counties with more highly levered firms experience larger county-level employment declines in response to the same local house-price declines. Hence, the mechanism is not undone by general-equilibrium reallocation across firms within a county.
- **Why the paper matters.** The paper inserts firms into the household-demand narrative of the Great Recession. It shows that households did not mechanically "cause" layoffs by reducing spending; rather, weak firm balance sheets shaped how consumer-demand shocks translated into employment losses.

2 Data and measurement (key objects)

2.1 Observed establishment-level panel

- **Unit of observation.** The main unit is the establishment: a physical operating location such as a retail store, restaurant, warehouse, or plant.

- **Employment data.** Establishment employment comes from the U.S. Census Bureau's Longitudinal Business Database (LBD), which covers essentially all U.S. business establishments with at least one paid employee.
- **Firm balance sheets.** Parent-firm balance sheet and income statement data come from Compustat. Financial firms and utilities are excluded.
- **House prices.** Local house prices come from Zillow at the zip-code level when available and county level otherwise.
- **Matched sample size.** The match between Compustat firms and LBD establishments yields 327,500 establishments with employment data from 2007 to 2009. After matching house prices, the final estimation sample contains 284,800 establishments.

2.2 Observed firm-level and county-level data

- **Parent firms.** The firm-level sample contains about 2,800 parent firms. These are used to measure leverage, profitability, and other balance-sheet variables.
- **County aggregation.** The county-level analysis is based on 1,000 counties. County employment is aggregated either over all matched Compustat-LBD firms or, in an expanded exercise, over all firms in the full LBD.
- **County controls.** The paper includes county-by-industry employment share controls based on 23 two-digit NAICS industries to keep county industrial composition fixed in aggregate regressions.

2.3 Key measured objects

- **Leverage.**

  $$\text{Leverage}_{f,06} = \frac{\text{debt in current liabilities}_{f,06} + \text{long-term debt}_{f,06}}{\text{total assets}_{f,06}}.$$

  It is measured at the parent-firm level in 2006, before the recessionary employment losses.
- **Employment change.**

  $$\Delta \log Emp_{e,07-09} = \log(Emp_{e,2009}) - \log(Emp_{e,2007}),$$

  the percentage change in establishment-level employment from 2007 to 2009.
- **House-price change.**

  $$\Delta \log HP_{z,06-09} = \log(HP_{z,2009}) - \log(HP_{z,2006}),$$

  where z is the establishment's zip code or county.
- **Alternative balance-sheet measures.** The paper also uses net leverage, market leverage, debt-to-EBITDA, interest coverage, and the Whited–Wu and Kaplan–Zingales financial-constraints indexes.
- **Firm-level outcomes.** To examine mechanism, the paper studies short-term debt issuance, long-term debt issuance, equity issuance, establishment closures, and capital expenditures.

- **Industry classification.** Establishments are split into tradable, nontradable, and "other" industries. The nontradable/tradable classification follows Mian and Sufi (2014a).

2.4 Unobservables and timing

- **Timing.** Leverage is measured in 2006, house-price changes over 2006–2009, and employment changes over 2007–2009. Thus, the paper interacts a pre-crisis balance-sheet variable with a recession-period demand shock.
- **Main unobservable.** The key omitted variable concern is local economic weakness that simultaneously drives house prices and employment. The paper addresses this with fixed effects, placebo sectors, and an IV for house prices.
- **Pre-crisis differences.** High-leverage firms are somewhat less profitable and expanded somewhat more before the crisis, so the paper explicitly tests those alternative stories.
- **Useful percentiles.** The 10th and 90th percentiles used in the paper's counterfactuals are:

$$\Delta \log HP^{10}_{06-09} = -0.402, \qquad \Delta \log HP^{90}_{06-09} = 0.035,$$

$$Leverage^{10}_{06} = 0.000, \qquad Leverage^{90}_{06} = 0.568.$$

2.5 Empirical applications

The paper's empirical strategy has four connected applications:

- establishment-level employment responses,
- extensive-margin closures,
- firm-level financing and investment responses,
- county-level employment responses and policy interpretation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework

The paper's mechanism can be summarized as

$$\text{house price decline} \Rightarrow \text{lower local consumer demand} \Rightarrow \text{lower firm revenue}.$$

With no financial frictions, firms could smooth through this shock and hoard labor. With weak balance sheets, however, firms have difficulty financing temporary wage bills and respond with sharper layoffs and closures.

The paper is therefore not a standard financial-accelerator story driven by aggregate net-worth shocks. Instead, it studies an interaction between *heterogeneous local demand shocks* and *heterogeneous firm balance sheets.*

3.2 Baseline establishment-level employment regression

The core specification is

$$\Delta \log Emp_e = \beta \, \Delta \log HP_z + \gamma \left(\Delta \log HP_z \times Leverage_f\right) + \delta \, Leverage_f + \text{FE} + u_e.$$

Here:

- e indexes establishments,
- f indexes the parent firm,
- z indexes the local housing market,
- $\Delta \log HP_z$ captures the local consumer-demand shock,
- γ is the coefficient of interest.

Interpretation.

- β is the employment elasticity with respect to local house prices for a zero-leverage firm.
- γ tells us how much stronger that elasticity becomes as leverage rises.
- A positive γ means that higher leverage amplifies the employment response to local demand shocks.

3.3 Counterfactual elasticities

Given the linear interaction, the elasticity at leverage level L is

$$\frac{\partial \Delta \log Emp_e}{\partial \Delta \log HP_z} = \beta + \gamma L.$$

Using the paper's preferred baseline numbers $\beta \approx 0.029$ and $\gamma \approx 0.114$, the implied elasticities are

$$\beta + \gamma L^{10} = 0.029 + 0.114 \times 0.000 \approx 0.029,$$

$$\beta + \gamma L^{90} = 0.029 + 0.114 \times 0.568 \approx 0.094.$$

So an establishment at the 90th percentile of leverage has an employment elasticity with respect to house prices that is roughly three times that of an establishment at the 10th percentile.

Interpretation. The same fall in local demand causes much larger employment losses when the parent firm starts the crisis with a weaker balance sheet.

3.4 Fixed effects and preferred specification

The paper estimates several variants of the baseline regression with different fixed effects:

- no fixed effects,
- industry fixed effects,

- zip-code fixed effects,
- firm fixed effects,
- zip-code × industry fixed effects.

The "tightest" specification compares establishments in the same zip code and four-digit NAICS industry, but the authors emphasize that adding firm fixed effects may overcontrol if firms are geographically concentrated or if internal capital markets partly smooth shocks across their own establishments.

Interpretation. The stability of the interaction coefficient across many fixed-effect structures is important because it shows the result is not driven by industry composition or broad regional sorting.

3.5 Alternative balance-sheet measures

The paper replaces leverage with other debt-capacity or balance-sheet-strength measures:

$$NetLeverage_{06}, \quad MarketLeverage_{06}, \quad Debt/EBITDA_{06}, \quad InterestCoverage_{06}, \quad \Delta Leverage_{02-06}, \quad WW_{06}, \quad KZ_{06}.$$

The regression becomes

$$\Delta \log Emp_e = \beta\, \Delta \log HP_z + \gamma\, (\Delta \log HP_z \times DebtCapacity_f) + \delta\, DebtCapacity_f + \mathrm{FE} + u_e.$$

Interpretation. If the balance-sheet mechanism is real, it should not hinge on a single accounting definition of leverage. The paper finds exactly this robustness.

3.6 Industry-sector regressions

To distinguish local demand from other channels, the paper estimates the same employment regression separately for nontradable and tradable sectors:

$$\Delta \log Emp_e^{(s)} = \beta_s\, \Delta \log HP_z + \gamma_s\, (\Delta \log HP_z \times Leverage_f) + \delta_s\, Leverage_f + \mathrm{FE} + u_e,$$

where $s \in \{\text{nontradable}, \text{tradable}\}$.

Interpretation.

- In nontradables, local spending matters directly, so one should see a strong house-price/employment relationship.
- In tradables, local house prices should matter much less because demand is national or global.
- The tradable sector therefore serves as a placebo test for the consumer-demand channel.

3.7 Establishment closures

To examine the extensive margin, the paper replaces the dependent variable with a closure indicator:

$$Closure_e = \beta\, \Delta \log HP_z + \gamma\, (\Delta \log HP_z \times Leverage_f) + \delta\, Leverage_f + \mathrm{FE} + u_e.$$

Interpretation. A positive interaction in absolute value here means highly levered firms not only cut jobs more, but also shut down establishments more readily when local demand collapses.

3.8 Firm-level financing and investment regressions

To test whether high-leverage firms *act* like constrained firms in the recession, the paper aggregates local house-price shocks to the firm level and estimates

$$Y_f = \beta \, \Delta \log HP_f + \gamma \, (\Delta \log HP_f \times Leverage_f) + \delta \, Leverage_f + \text{industry FE} + u_f,$$

where Y_f is one of:

- short-term debt growth,
- long-term debt growth,
- equity issuance,
- employment growth,
- establishment closure rate,
- capital expenditures relative to PP&E.

Interpretation.

- If high-leverage firms are financially constrained, they should be less able to raise debt in bad times.
- If they cannot smooth financially, they should cut labor more, close more establishments, and reduce investment more.

3.9 County-level aggregate regressions

To test whether the balance-sheet channel survives aggregation, the paper estimates

$$\Delta \log Emp_c = \beta \, \Delta \log HP_c + \gamma \, (\Delta \log HP_c \times Leverage_c) + \delta \, Leverage_c + X_c' \theta + u_c,$$

where:

- c indexes counties,
- county leverage is the employment-weighted average leverage of firms in the county,
- X_c contains county-by-industry employment shares.

Interpretation.

- If workers and output simply reallocated frictionlessly from weak to strong firms, the effect of leverage might disappear in aggregate.
- A positive and significant γ at the county level means local balance-sheet composition matters for aggregate employment losses.

3.10 Instrumenting house-price changes

A major concern is that house-price declines may proxy for other local shocks. The paper therefore instruments $\Delta \log HP$ using Saiz's (2010) housing supply elasticity:

$$\Delta \log HP_z = \pi \, HSE_z + \text{controls} + v_z,$$

where HSE_z captures local geographical and regulatory constraints on new construction.

The second stage then uses predicted house-price changes inside the employment regressions above.

Interpretation. Places with more inelastic housing supply saw larger house-price movements in the crash for quasi-exogenous reasons. If the leverage interaction survives IV, it is harder to attribute the results to omitted local shocks.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

A naive reading of the correlation between leverage and employment losses would be unconvincing for several reasons.

- Highly levered firms may simply be low-quality firms.
- They may have expanded too aggressively during the boom.
- They may be concentrated in sectors or regions hit harder in the recession.
- They may have activist owners or high betas that make them intrinsically more cyclical.

The paper's design is built to rule out these alternative stories one by one.

4.2 Why the main interaction is informative

The key source of identification is that parent-firm leverage in 2006 is essentially uncorrelated with local house-price changes during the Great Recession. In the data, low- and high-leverage establishments face similar local demand shocks, but react differently.

This means the regression does not compare high-leverage firms in deeply collapsing regions to low-leverage firms in stable regions. Rather, it compares how different firms respond to similar local demand contractions.

4.3 Fixed-effect strategy

The fixed-effect designs greatly strengthen the argument.

- **Industry fixed effects** absorb broad sectoral differences.
- **Zip-code fixed effects** absorb local average shocks.

- **Zip-code $\times$ industry fixed effects** compare establishments in the same local market and same narrow industry.
- **Firm fixed effects** compare different establishments within the same firm, though these may overcontrol.

Under the zip-code $\times$ industry specification, the identifying variation comes from establishments selling similar goods in the same local demand environment but belonging to firms with different balance-sheet strength.

4.4 Tradable-sector placebo

The paper's most transparent placebo is the tradable sector. If falling house prices were merely a proxy for some broader local shock, one might expect tradable employment to move similarly. But the paper finds essentially no significant house-price/employment relationship in tradables.

This is powerful because it aligns exactly with the consumer-demand interpretation: local house-price declines should hit locally consumed goods more than nationally traded goods.

4.5 Instrument for house prices

The IV based on housing supply elasticity addresses the concern that employment and house prices are jointly driven by local fundamentals.

The logic is standard:

- inelastic local housing supply creates larger price swings,
- this source of variation is plausibly unrelated to recession-period establishment employment conditional on controls,
- the second-stage leverage interaction remains and is, if anything, slightly stronger.

The paper also uses only the topographical part of the instrument (share of unavailable land) to reduce concern that regulatory housing restrictions are themselves endogenous.

4.6 Alternative-channel tests

The paper then runs a large battery of horse races against competing explanations.

- **Overexpansion.** Control for employment growth and asset growth from 2002 to 2006.
- **Low productivity.** Control for ROA, net profit margin, and TFP.
- **High beta / cyclical sensitivity.** Control for historical sensitivities of employment and house prices.
- **Activist ownership.** Control for hedge fund and private equity involvement.

In all cases, the leverage interaction remains stable and significant.

4.7 General-equilibrium concern

A natural concern is that weak firms may lay off workers but strong firms in the same county may hire them, undoing the effect in aggregate. The county-level results reject this interpretation. Even when the outcome is county-level employment, higher county leverage amplifies employment losses following local house-price declines.

The paper further expands the outcome to employment of *all firms* in the LBD, not just matched public firms, and still finds a significant interaction.

4.8 Relation to the literature

The paper connects three literatures:

- household-debt and demand-shock studies of the Great Recession,
- firm balance-sheet and financial-frictions models,
- labor-hoarding theories.

Its contribution is to connect them empirically: local demand shocks matter much more for employment when they hit weak corporate balance sheets.

5 Tables and figures

5.1 Table I: Summary statistics and leverage correlations

Read:

- The average parent-firm leverage in 2006 is 0.227, with substantial dispersion.
- The 10th and 90th percentiles of leverage are 0.000 and 0.568, respectively.
- The 10th and 90th percentiles of the local house-price change are -0.402 and 0.035, so the housing shock is economically large.
- High-leverage firms are less profitable on average and more financially constrained according to WW and KZ, which motivates the mechanism but also creates alternative-story concerns the paper must address.

5.2 Figure I: House prices, employment, and leverage quartiles

Read:

- The graph provides the main visual fact of the paper.
- In the lowest leverage quartile, the employment elasticity with respect to house prices is only about 0.020.
- In the highest leverage quartile, the elasticity rises to about 0.096.

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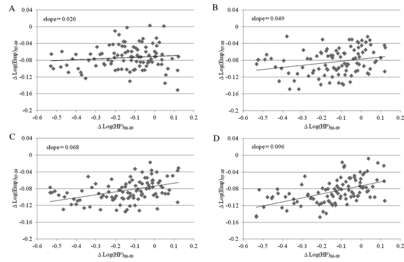

FIGURE I

Firm Leverage, House Prices, and Employment at the Establishment Level

The figure plots the percentage change in establishment-level employment between 2007 and 2009, ΔLog(Emp)$_{07-09}$, against the percentage change in house prices in the establishment's zip code or county between 2006 and 2009, ΔLog(HP)$_{06-09}$, for different quartiles of firm leverage. For each percentile of ΔLog(HP)$_{06-09}$, the scatterplot depicts the mean values of ΔLog(HP)$_{06-09}$ and ΔLog(Emp)$_{07-09}$, respectively.

- So before any regression machinery, the basic pattern already suggests that leverage amplifies local employment responses.

5.3 Table II: Baseline establishment-level regressions

Read:

- The average elasticity of establishment employment with respect to local house prices is about 0.066 in the simplest specification.
- The interaction coefficient $\Delta \log HP \times Leverage$ is around 0.11 in the main baseline regressions and remains positive and significant even in much tighter specifications.
- The coefficient is still about 0.075 in the zip-code $\times$ industry plus firm fixed-effect specification.
- This table is the empirical core: local demand shocks hurt employment more at more highly levered firms.

TABLE II

FIRM LEVERAGE, CONSUMER DEMAND, AND EMPLOYMENT

	ΔLog(Emp)$_{07-09}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔLog(HP)$_{06-09}$	0.066***	0.029	0.029		0.027		
	(0.019)	(0.022)	(0.019)		(0.019)		
ΔLog(HP)$_{06-09}$ $\times$ Leverage$_{06}$		0.111***	0.114***	0.113***	0.084**	0.076**	0.075**
		(0.039)	(0.040)	(0.038)	(0.035)	(0.031)	(0.038)
Leverage$_{06}$		−0.028**	−0.032**	−0.020**			
		(0.014)	(0.015)	(0.009)			
Industry fixed effects	No	No	Yes	Yes	Yes	Yes	—
Firm fixed effects	No	No	No	No	Yes	Yes	Yes
Zip code fixed effects	No	No	No	Yes	No	Yes	—
Zip code $\times$ industry fixed effects	No	No	No	No	No	No	Yes
R-squared	0.00	0.00	0.04	0.13	0.17	0.25	0.31
Observations	284,800	284,800	284,800	284,800	284,800	284,800	284,800

Notes. The dependent variable, ΔLog(Emp)$_{07-09}$, is the percentage change in establishment-level employment from 2007 to 2009. ΔLog(HP)$_{06-09}$ is the percentage change in house prices in the establishment's zip code or county (if the zip code information is missing) from 2006 to 2009. Leverage$_{06}$ is the ratio of the sum of debt in current liabilities and long-term debt to total assets associated with the establishment's parent firm in 2006. Industry fixed effects are based on four-digit NAICS codes. All regressions are weighted by establishment size. Standard errors (in parentheses) are clustered at both the state and firm level. **, and *** denote significance at the 5%, and 1% levels, respectively.

TABLE III

OTHER MEASURES OF FIRMS' BALANCE SHEET STRENGTH

	ΔLog(Emp)$_{07-09}$						
	Net leverage$_{06}$ (1)	Market leverage$_{06}$ (2)	Debt to EBITDA$_{06}$ (3)	Interest coverage$_{06}$ (4)	ΔLeverage$_{02-06}$ (5)	WW$_{06}$ (6)	KZ$_{06}$ (7)
ΔLog(HP)$_{06-09}$	0.029	0.032	0.036*	0.040*	0.011	0.027	0.029
	(0.020)	(0.019)	(0.019)	(0.020)	(0.033)	(0.023)	(0.021)
ΔLog(HP)$_{06-09}$ $\times$ Debt capacity	0.120***	0.130***	0.012***	0.127**	0.223***	0.059***	0.003**
	(0.041)	(0.044)	(0.004)	(0.056)	(0.070)	(0.015)	(0.002)
Debt capacity	−0.038**	−0.054***	−0.003**	−0.063***	−0.038*	−0.009**	−0.003***
	(0.016)	(0.018)	(0.001)	(0.024)	(0.022)	(0.004)	(0.000)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Observations	284,800	284,800	284,800	284,800	284,800	284,800	284,800

Notes. This table presents variants of the regressions in Table II in which Leverage$_{06}$ is replaced by other measures of firms' debt capacity or balance sheet strength. Net leverage is the ratio of debt in current liabilities plus long-term debt minus cash and short-term investments divided by total assets. Market leverage is the ratio of debt in current liabilities plus long-term debt divided by total assets minus the book value of equity plus the market value of equity (stock price multiplied by the number of shares outstanding). Debt to EBITDA is the ratio of debt in current liabilities plus long-term debt divided by operating income before depreciation. Interest coverage is the ratio of interest expense to operating income after depreciation. ΔLeverage$_{02-06}$ is the change in firm leverage from 2002 to 2006. WW and KZ are the financial constraints indexes of Whited and Wu (2006) and Kaplan and Zingales (1997), respectively. ΔLeverage, WW, and KZ are net of their respective minimum values. Industry fixed effects are based on four-digit NAICS codes. All regressions are weighted by establishment size. Standard errors (in parentheses) are clustered at both the state and firm level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.4 Table III: Other measures of firms' balance-sheet strength

Read:

- The result is not special to one leverage measure.
- Net leverage, market leverage, debt to EBITDA, interest coverage, leverage growth from 2002 to 2006, and the WW and KZ indexes all tell a similar story.
- In particular, the positive interaction between the demand shock and debt-capacity weakness survives throughout.
- This table makes the interpretation more structural: what matters is not accounting leverage per se, but weak debt capacity and tight financing conditions.

5.5 Table IV and Figure II: Nontradable versus tradable industries

TABLE IV
TRADABLE AND NONTRADABLE INDUSTRIES

	ΔLog(Emp)$_{07-09}$			
	(1) Nontradable	(2) Tradable	(3) Nontradable	(4) Tradable
ΔLog(HP)$_{06-09}$	0.074**	0.009	0.029	−0.015
	(0.035)	(0.019)	(0.019)	(0.043)
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$			0.131***	0.037
			(0.034)	(0.120)
Leverage$_{06}$			−0.038**	−0.026
			(0.015)	(0.020)
Industry fixed effects	Yes	Yes	Yes	Yes
R-squared	0.04	0.03	0.04	0.03
Observations	124,100	9,900	124,100	9,900

Notes. This table presents variants of the regressions in Table II in which the sample is restricted to tradable and nontradable industries, respectively. Tradable and nontradable industries are described in Mian and Sufi (2014a). All regressions are weighted by establishment size. Standard errors (in parentheses) are clustered at both the state and firm level. **, and *** denote significance at the 5%, and 1% levels, respectively.

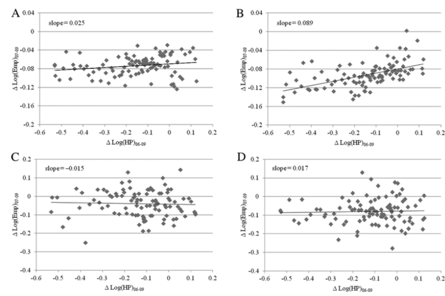

FIGURE II
Nontradable and Tradable Employment at the Establishment Level

The plots are similar to those in Panels A and D of Figure I, except that the sample is restricted to nontradable and tradable industries, respectively.

Read:

- In the nontradable sector, the house-price elasticity of employment is about 0.074 and the interaction with leverage is about 0.131.
- In the tradable sector, both the main house-price effect and the interaction are small and insignificant.
- The "other" sector looks very similar to nontradables, and together these sectors account for 97% of establishment observations.
- This is one of the cleanest pieces of evidence that the mechanism is local consumer demand rather than generic recession sensitivity.

TABLE V
FIRM-LEVEL ANALYSIS

	External finance			Employment and investment		
	ΔST debt$_{07-09}$ (1)	ΔLT debt$_{07-09}$ (2)	ΔEquity$_{07-09}$ (3)	ΔLog(Emp)$_{07-09}$ (4)	Est. Closure$_{07-09}$ (5)	ΔCAPEX$_{07-09}$ (6)
ΔLog(HP)$_{06-09}$	−0.025**	−0.040**	0.005	0.020	−0.008	0.002
	(0.011)	(0.019)	(0.037)	(0.033)	(0.015)	(0.005)
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$	0.035**	0.059**	−0.011	0.122***	−0.046**	0.014**
	(0.014)	(0.021)	(0.047)	(0.040)	(0.019)	(0.007)
Leverage$_{06}$	−0.011*	−0.019*	0.009	−0.024**	0.018**	−0.005*
	(0.006)	(0.011)	(0.021)	(0.010)	(0.008)	(0.003)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.09	0.08	0.03	0.11	0.11	0.14
Observations	2,800	2,800	2,800	2,800	2,800	2,800

Notes. This table presents firm-level variants of the regressions in Table II. Short-term (ST) debt is the ratio of debt in current liabilities divided by total assets. Long-term (LT) debt is the ratio of long-term debt divided by total assets. Equity is the ratio of the book value of equity divided by total assets. Establishment (Est.) closure is the number of establishments closed between 2007 and 2009 divided by the number of establishments in 2007. CAPEX is the ratio of capital expenditures to property, plant and equipment (PP&E). ΔLog(HP)$_{06-09}$ is aggregated at the firm level by computing the employment-weighted average value of Δ Log(HP)$_{06-09}$ across all of the firm's establishments. Industry fixed effects are based on four-digit NAICS codes. Standard errors (in parentheses) are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.6 Table V: High-leverage firms behave like financially constrained firms

Read:

- When local demand falls, high-leverage firms are less able to raise short-term and long-term debt.
- They cut employment more, close establishments more, and reduce capital expenditures more.
- Equity issuance does not offset the funding shortfall in a meaningful way.
- This is the mechanism table: high leverage is not only correlated with bad outcomes, it predicts the financing behavior one would expect from constrained firms.

TABLE VI

ROBUSTNESS

	(1)	(2)	(3)	(4)
Panel A: Employment and asset growth				
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$	0.111**	0.113***		
	(0.047)	(0.040)		
ΔLog(HP)$_{06-09}$ × Δ Log(Emp)$_{02-06}$	0.027			
	(0.034)			
ΔLog(HP)$_{06-09}$ × Δ Log(Assets)$_{02-06}$		0.012		
		(0.009)		
Panel B: Productivity				
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$	0.101**	0.122***	0.108***	
	(0.040)	(0.041)	(0.039)	
ΔLog(HP)$_{06-09}$ × ROA$_{06}$	−0.024			
	(0.110)			
ΔLog(HP)$_{06-09}$ × NPM$_{06}$		−0.038		
		(0.161)		
ΔLog(HP)$_{06-09}$ × TFP$_{06}$			−0.049*	
			(0.026)	
Panel C: Sensitivity to aggregate employment and house prices				
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$	0.109**	0.110**	0.119***	0.120***
	(0.044)	(0.044)	(0.044)	(0.044)
ΔLog(HP)$_{06-09}$ × Elasticity$_{\text{Emp,10-year}}$	0.006*			
	(0.004)			
ΔLog(HP)$_{06-09}$ × Elasticity$_{\text{Emp,20-year}}$		0.005		
		(0.004)		
ΔLog(HP)$_{06-09}$ × Elasticity$_{\text{HP, 10-year}}$			0.006	
			(0.008)	
ΔLog(HP)$_{06-09}$ × Elasticity$_{\text{HP, 02-06}}$				0.007
				(0.008)

TABLE VI

(CONTINUED)

	(1)	(2)	(3)	(4)
Panel D: Activist investors				
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$	0.110***	0.115***		
	(0.039)	(0.040)		
ΔLog(HP)$_{06-09}$ × Hedge Fund$_{06}$	0.036			
	(0.036)			
ΔLog(HP)$_{06-09}$ × Private Equity$_{06}$		0.009		
		(0.056)		

Notes. This table presents variants of the regressions in Table II in which Z and ΔLog(HP)06–09 × Z are included as additional controls. For brevity, the table only shows the coefficients associated with the main variables of interest, ΔLog(HP)06–09 × Leverage06 and ΔLog(HP)06–09 × Z. The full sets of coefficients are provided in Tables 10 to 14 of the Online Appendix. In Panel A, Z is either the growth in firm-level employment from 2002 to 2006, Δ Log(Emp)$_{02-06}$ (column (1)), or the growth in firm-level assets from 2002 to 2006, ΔLog(Assets)$_{02-06}$ (column (2)). In Panel B, Z is either the firm's return on assets, ROA$_{06}$ (column (1)), the firm's net profit margin, NPM$_{06}$ (column (2)), or the firm's total factor productivity, TFP$_{06}$ (column (3)), all measured in 2006. ROA$_{06}$ is the ratio of operating income before depreciation to total assets, NPM$_{06}$is the ratio of operating income before depreciation to sales, and TFP$_{06}$ is the residual from a regression of Log(Sales) on Log(Employees) and Log(PP&E) across all Compustat firms in the same two-digit SIC industry. In Panel C, Z is either the 10-year elasticity of firm-level employment with respect to aggregate employment, Elasticity$_{Emp,10\text{-}year}$(column (1)), the 20-year elasticity of firm-level employment with respect to aggregate employment, Elasticity$_{Emp,20\text{-}year}$ (column (2)), the 10-year elasticity of firm-level employment with respect to house prices, Elasticity$_{HP,10\text{-}year}$ (column (3)), or the elasticity of firm-level employment with respect to house prices during the 2002 to 2006 housing boom, Elasticity$_{HP,02-06}$ (column (4)). Elasticity$_{Emp,10\text{-}year}$ and Elasticity$_{Emp,20\text{-}year}$ are computed by estimating a firm-level regression of ΔLog (Employment) on a constant and ΔLog(LBD Employment) using all available years from 1997 to 2006 and 1987 to 2006, respectively. Elasticity$_{HP,10\text{-}year}$ and Elasticity$_{HP,02-06}$ are computed as the employment-weighted average elasticity of employment with respect to house prices across all of the firm's establishments, where the latter is computed either by estimating an establishment-level regression of ΔLog(Employment) on a constant and ΔLog(HP) using all available years from 1997 to 2006 (Elasticity$_{HP,10\text{-}year}$) or as the percentage change in establishment-level employment divided by the percentage change in house prices during the 2002 to 2006 housing boom (Elasticity$_{HP,02-06}$). In Panel D, Z is either a dummy indicating whether a firm is targeted by an activist hedge fund, Hedge Fund$_{06}$ (column (1)), or a dummy indicating whether a firm has significant (i.e., more than 5%) private equity ownership, Private Equity$_{06}$ (column (2)), all measured in 2006. Industry fixed effects are based on four-digit NAICS codes. All regressions are weighted by establishment size. Standard errors (in parentheses) are clustered at both the state and firm level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.7 Table VI: Robustness against alternative stories

Read:

- The leverage interaction remains significant after controlling for pre-crisis employment growth and asset growth, so the effect is not driven by firms that "expanded too much."
- It also survives controls for ROA, profit margins, and TFP, so it is not just low-productivity firms reacting more.
- It survives controls for historical sensitivities of employment and house prices and for activist investors.
- This table does the heavy lifting in showing that the paper is really about balance sheets, not about a bundle of unrelated bad-firm characteristics.

5.8 Figure III and Table VII: County-level employment

Read:

- The county-level elasticity of employment with respect to house prices is about 0.069 overall and 0.076 in nontradables.
- The interaction between county leverage and house-price changes is about 0.110 overall and 0.134 in nontradables.

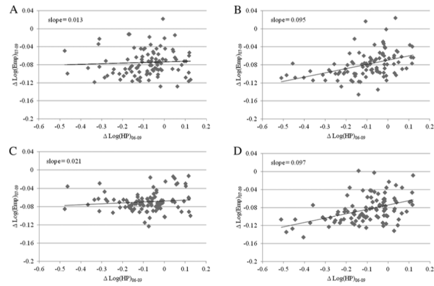

FIGURE III

Firm Leverage, House Prices, and Employment at the County Level

The plots are similar to those in Panels A and D of Figure I and Panels A to D in Figure II, except that the analysis is at the county level. County-level employment is total employment by all firms in our sample within a county. County-level leverage is the employment-weighted average value of firm leverage across all firms in our sample within a county.

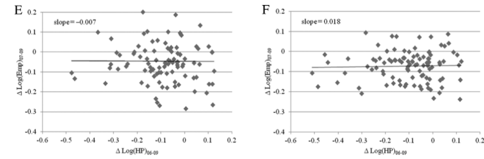

FIGURE III

(CONTINUED)

TABLE VII

COUNTY-LEVEL ANALYSIS

	ΔLog(Emp)$_{07-09}$					
	All industries		Nontradable		Tradable	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔLog(HP)$_{06-09}$	0.069***	0.025	0.076***	0.026	0.010	−0.015
	(0.008)	(0.019)	(0.012)	(0.020)	(0.010)	(0.039)
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$		0.110***		0.134***		0.034
		(0.038)		(0.036)		(0.151)
Leverage$_{06}$		−0.029**		−0.039**		−0.028*
		(0.013)		(0.020)		(0.015)
County × industry controls	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.13	0.13	0.18	0.19	0.09	0.09
Observations	1,000	1,000	1,000	1,000	1,000	1,000

Notes. This table presents county-level variants of the regressions in Table II and IV. County-level employment is total employment by all firms in our sample within a county. County-level leverage is the employment-weighted average value of Leverage$_{06}$ across all firms in our sample within a county. County × industry controls are the county-specific employment shares of all 23 two-digit NAICS industries in 2006. All regressions are weighted by county size. Standard errors (in parentheses) are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

- In the highest county-leverage quartile, the county-level employment elasticity is about 0.095, versus a very small elasticity in the lowest quartile.
- So the composition of firm leverage matters for aggregate local employment outcomes, not just for within-county redistribution across firms.

5.9 Table VIII: Expanding to all firms in the LBD

Read:

- When county employment is measured using *all* firms in the LBD, the average house-price elasticity rises to about 0.104, because smaller firms are even more financially constrained.
- Even though county leverage is now measured with noise, the leverage interaction remains positive and significant.
- The interaction is about 0.045 overall and 0.058 in nontradables.
- This is an important external-validity check: the balance-sheet channel is not just a public-firm phenomenon.

5.10 Extensive-margin and industry-detail evidence

Read:

- Falling house prices increase establishment closures, especially for more highly levered firms.
- The industries most exposed to house-price declines are local-demand sectors such as restaurants, building-material retailers, health and personal care stores, and several auto-related activities.

TABLE VIII

EXPANDING THE SAMPLE: COUNTY-LEVEL EMPLOYMENT BY ALL FIRMS IN THE LBD

	ΔLog(Emp)$_{07-09}$					
	All industries		Nontradable		Tradable	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔLog(HP)$_{06-09}$	0.104***	0.085***	0.122***	0.096***	0.008	−0.003
	(0.005)	(0.015)	(0.006)	(0.019)	(0.016)	(0.034)
ΔLog(HP)$_{06-09}$ × Leverage$_{06}$		0.045**		0.058***		0.022
		(0.022)		(0.020)		(0.105)
Leverage$_{06}$		−0.023**		−0.028**		−0.019*
		(0.010)		(0.014)		(0.010)
County × industry controls	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.14	0.14	0.16	0.16	0.13	0.13
Observations	1,000	1,000	1,000	1,000	1,000	1,000

Notes. This table presents variants of the regressions in Table VII in which county-level employment is total employment by all firms in the LBD within a county. County-level leverage is the employment-weighted average value of Leverage$_{06}$ across all firms in our sample within a county. County × industry controls are the county-specific employment shares of all 23 two-digit NAICS industries in 2006. All regressions are weighted by county size. Standard errors (in parentheses) are clustered at the state level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

- No tradable industry appears among the sectors most sensitive to local house prices.

5.11 Policy discussion: labor hoarding and work sharing

Read:

- The policy discussion is not a table, but it is central to the paper's interpretation.
- The authors argue that financially constrained firms may be unable to hoard labor efficiently during temporary demand collapses.
- Germany's short-time work system is presented as an example of firm-targeted employment support that may have preserved matches by subsidizing hour reductions instead of layoffs.
- The paper is careful, however, to stress that such subsidies should be temporary relief, not permanent protection.

6 Short summary

- **Conclusion.** Local consumer-demand shocks translated into much larger employment losses when they hit firms with weak balance sheets.
- **Identification insight.** The paper identifies the balance-sheet channel by comparing establishments facing similar local house-price declines but belonging to firms with different leverage, and by showing that the effect appears in nontradables but not tradables.
- **Empirical lesson.** Highly levered firms do not merely look weak ex ante; in the recession they behave like financially constrained firms by borrowing less, laying off more, closing more establishments, and investing less.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper establishes that firm balance sheets were an important propagation mechanism during the Great Recession. Establishments of more highly levered firms experienced significantly larger employment losses in response to local consumer-demand shocks.

- **Mechanism.** The results are consistent with financial constraints impairing labor hoarding. Weak firms cannot finance temporary payroll support when demand falls, so they respond with sharper layoffs and more closures.
- **Aggregate implication.** This is not just a micro composition effect. Counties with more highly levered firms also suffer larger employment losses, so leverage composition matters for aggregate local employment.
- **Credibility.** The paper's conclusions survive many demanding checks: rich fixed effects, IV for house prices, placebo tradable sectors, alternative leverage measures, and controls for productivity, pre-crisis expansion, and cyclical sensitivity.

7.2 Economic intuition

The economic intuition is simple but powerful.

A firm hit by a temporary negative demand shock would ideally like to retain workers if firing, rehiring, and retraining are costly. That is labor hoarding. But labor hoarding requires liquidity. A highly levered firm has less spare debt capacity, so the short-run cost of keeping workers looms larger. The same local demand shock therefore translates into a bigger employment cut.

At the same time, the paper shows why the household-demand story alone is incomplete. Households may cut spending when house prices fall, but firms with strong balance sheets can partially absorb that shock. Weak-balance-sheet firms cannot. The transmission from local demand to employment therefore depends crucially on corporate finance.

7.3 Takeaway

This paper is best read as a bridge between the household-debt view of the Great Recession and the corporate-finance view of labor adjustment. Its central message is that demand shocks and firm balance sheets interact. House-price declines hurt local demand, but they become especially destructive for employment when they hit firms that enter the downturn with high leverage and little financial slack. For work on recession propagation, labor hoarding, or firm-targeted stabilization policy, this paper is a benchmark reference.